# TH650A–CRB/TS2100

Industrial Robot

# INSTRUCTION MANUAL

## CLEAN TYPE
## INDUSTRIAL ROBOT SPECIFICATIONS

**Notice**

1. Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
2. Before operating the industrial robot, read through and completely understand this manual.
3. After reading through this manual, keep it nearby for future reference.

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Shibaura Machine

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The information contained in this manual is subject to change without prior notice to effect improvements.

# Preface

This manual describes the specifications of the TH–A series clean type industrial robot. This manual is essential to keep the robot performance for a long time, to prevent failures and to assure safety. Be sure to look through this manual and set up a maintenance program before actually starting the robot.

This manual is comprised of the following six (6) sections:

Section 1 Specifications
This section describes the basic specifications and names of respective parts of the clean type industrial robot.

Section 2 Transportation
This section describes how to remove the clean type robot from its box and how to transport it to the installation site.

Section 3 Installation
This section discusses the clean type robot installation environment, space requirements, and how to install the robot.

Section 4 Tool Interface
This section discusses how to connect the cables and pipelines for the tool of the clean type robot.

Section 5 Maintenance
This section describes the structure of the clean type robot and all items required for the maintenance and inspection of the same robot.

Section 6 Replacement Parts for Maintenance
This section explains the replacement parts for the maintenance.

# Precautions on Safety

Important information on the robot and controller is noted in the instruction manual to prevent injury to the user and persons nearby, prevent damage to assets and to ensure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

| Indication | Meaning of indication |
|---|---|
|  **DANGER** | This means that "incorrect handling will imminently lead to fatalities or major injuries". |
|  **WARNING** | This means that "incorrect handling may lead to fatalities or serious injuries." |
|  **CAUTION** | This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)". |

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long term treatment.

*2) Physical damage refers to major damage due to destruction of assets or resources.

[Explanation of symbols]

| Symbol | Meaning of symbol |
|---|---|
|  | This means that the action is prohibited (must not be done).<br>Details of the actions actually prohibited are indicated with pictures or words in or near the symbol. |
|  | This means that the action is mandatory (must must be done).<br>Details of the actions that must be done are indicated with pictures or words in or near the symbol. |
|  | This means danger.<br>Details of the actual danger are indicated with pictures or words in or near the symbol. |
|  | This means caution.<br>The details of the actual caution are indicated with pictures or words in or near the symbol. |

[Maintenance and inspection]

Be sure to observe the following items to use this product safely.

|  **DANGER** | |
|---|---|
|  <br>Prohibited | • DO NOT incinerate, disassemble or charge the batteries. Otherwise, they may rupture. |
|  <br>Mandatory | • Be sure to turn off the main power switch of the controller before starting inspection or maintenance.<br>• Batteries should be disposed of according to the user's in-house regulations. |

| ! CAUTION | |
|---|---|
|  Disassembly prohibited | • The user must NEVER replace or modify parts other than those described in the instruction manual. Otherwise, the performance may deteriorate or faults or accidents will be caused. |
|  Mandatory | • Always use the SHIBAURA MACHINE's designated spare parts when replacing the parts.<br>• Maintenance and inspection should be performed regularly. Otherwise, the system may malfunction or accidents will be caused. |

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# 1. Specifications

## 1.1 Name of Each Part

The names of respective parts of the clean type robot are shown in Fig. 1.1 below.

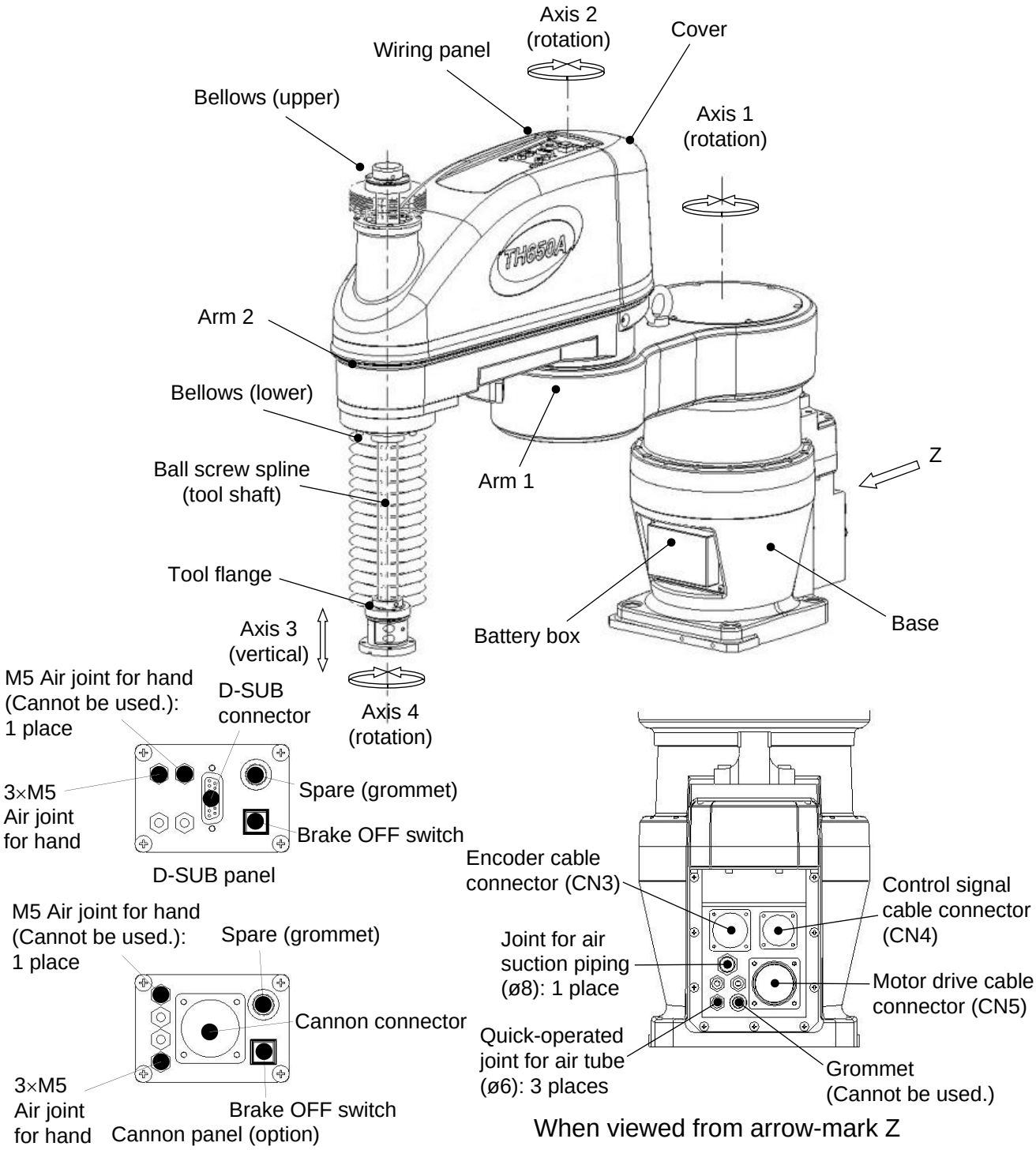



Fig. 1.1 Name of each part

## 1.2 Outer Dimensions

Fig. 1.2 and Fig. 1.3 show the outer dimensions and operating range of the robot.

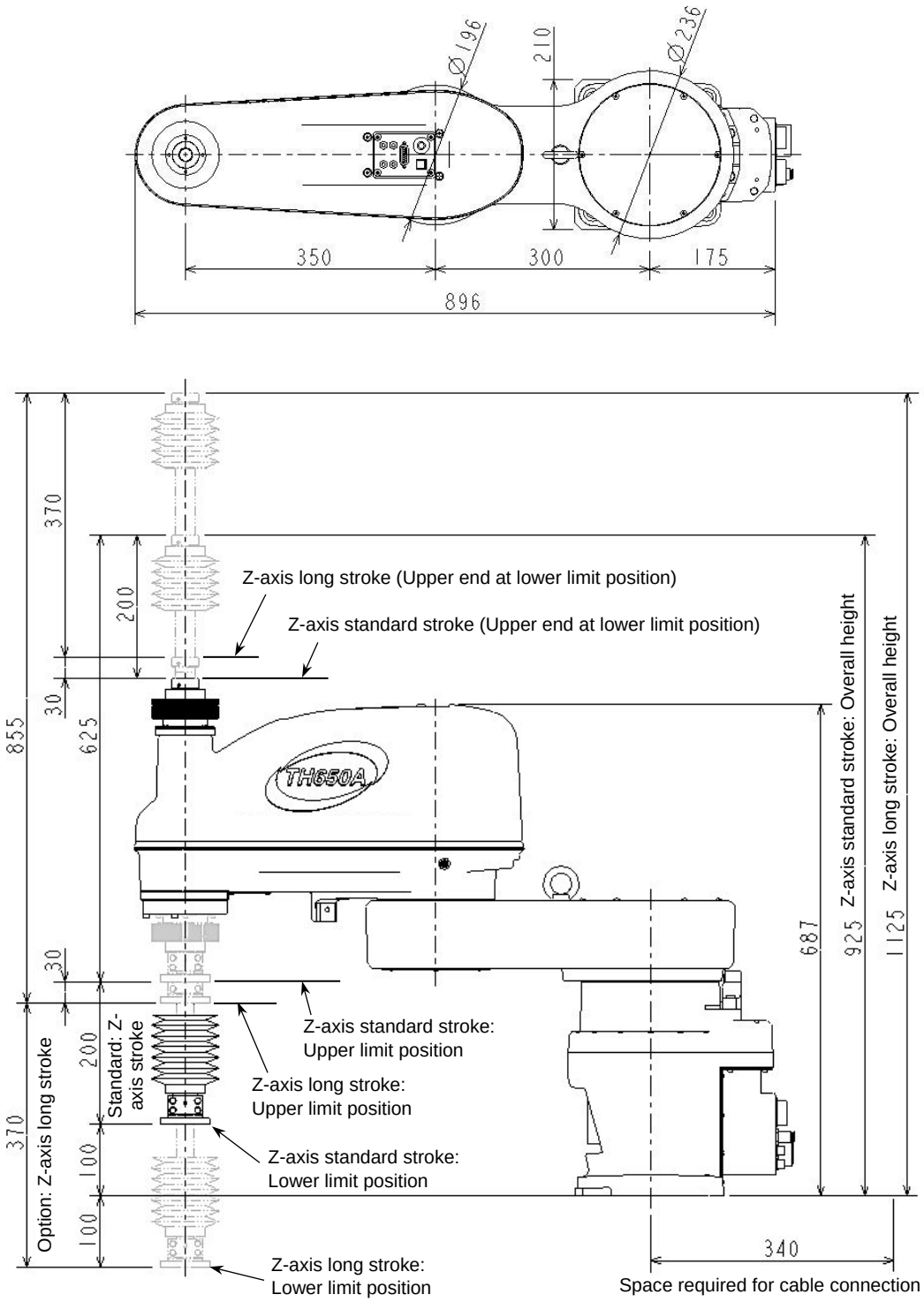



Fig. 1.2 Outer dimensions of the robot

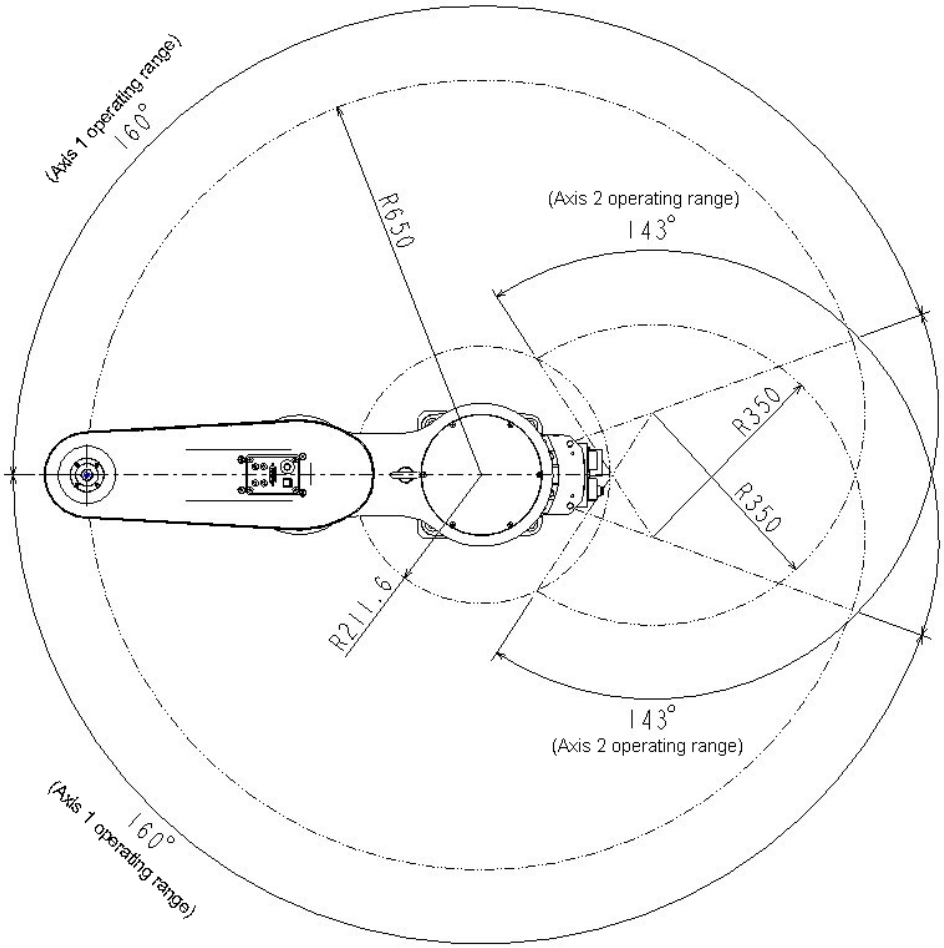



Fig. 1.3 Operating range of the robot

## 1.3 Specifications Table

| Item | | Specifications |
|---|---|---|
| Structure | | Horizontal multi-joint type SCARA robot |
| Model | | TH650A–CRB |
| Applicable controller | | TS2100 (*1) |
| Mass of robot body | | 53 kg |
| No. of controlled axes | | Four (4) |
| Arm length | | 650 mm (300+350) |
| Motor capacity | Axis 1 | 1,000 (W) |
| | Axis 2 | 600 (W) |
| | Axis 3 | 400 (W) |
| | Axis 4 | 400 (W) |
| Operating range | Axis 1 | ±160 (deg) |
| | Axis 2 | ±143 (deg) |
| | Axis 3 | 200 (mm) [Option: 370 (mm)] |
| | Axis 4 | ±360 (deg) |
| Maximum speed (*2) | Axis 1 | 340 (deg/s) |
| | Axis 2 | 600 (deg/s) |
| | Axis 3 | 2,050 (mm/s) |
| | Axis 4 | 1,700 (deg/s) |
| | Composite speed of axes 1 and 2 | 7.52 (m/s) |
| Rated payload mass | | 2 (kg) |
| Maximum payload mass | | 10 (kg) |
| Permissible load inertia (*2) | | 0.1 ($kg \cdot m^2$) |
| Repeatability (*3) | X, Y | ±0.01 (mm) |
| | Z | ±0.01 (mm) |
| | C | ±0.004 (deg) |
| Cycle time (*4) (When payload mass is 2 kg) | | 0.32 (sec) |
| Drive system | | By means of AC servo motors |
| Position detection method | | Absolute |

| Item | Specifications |
|---|---|
| Cleanness level (*5) | Ten (10) or less dust particles of 0.1 µm or over in diameter, which exist in 1 cft (28,317 $cm^3$) of a sample area. |
| Air suction volume | 60 normal liters/min |

*1: The structure of the robot controller is not of a clean type.

*2: When the mass of load exceeds 2 kg, or when the gravity center position of load is away from the axis 4 center position, both the speed and acceleration should be reduced, using the PAYLOAD command.

*3: The ambient temperature is predetermined at 20°C.

*4: Shuttle time for rough positioning in horizontal direction of 300 mm and vertical direction of 25 mm.

*5: When the air suction volume is 60 normal liters/min, down flowrate in the clean room is 0.4 m/s or over.

## 2. Transportation

### 2.1 Unpacking and Transport

The robot and controller are shipped either in clean bags (class 100 or equivalent), wooden crates or corrugated cartons according to the customer's requirements.
Open the packages in a location easily accessible, where the equipment is to be installed. Take careful precautions not to damage the robot and controller.
After opening the packages, make sure that all the accessories are present and that no part has been damaged during transport.
The package posture and contents of the wooden crates and corrugated cartons are the same as in the standard robot. For details, see the TH650A Installation and Transportation Manual provided separately.
The clean package condition is shown in Fig. 2.1.

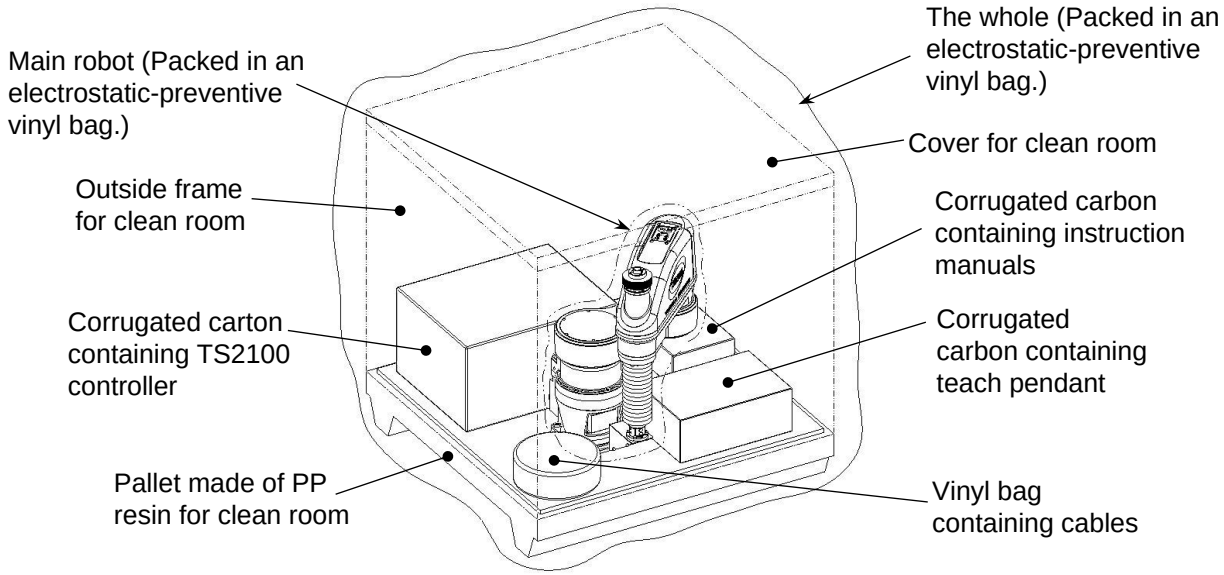



Fig. 2.1 Clean package condition

To unpack the shipment, observe the following procedures.

- Carry the shipment in a condition shown in Fig. 2.1 to the room just before the clean room, using a forklift truck, etc.
- Remove the outside electrostatic-preventive vinyl bag, cover and outside frame in the preceding room.
- Bring down all corrugated cartons containing the teach pendant, controller, etc.
- Remove the electrostatic-preventive vinyl bag containing the main robot.

- Carry the packages to the clean room, using a forklift truck for use in the clean room (which is to be provided by the customer).

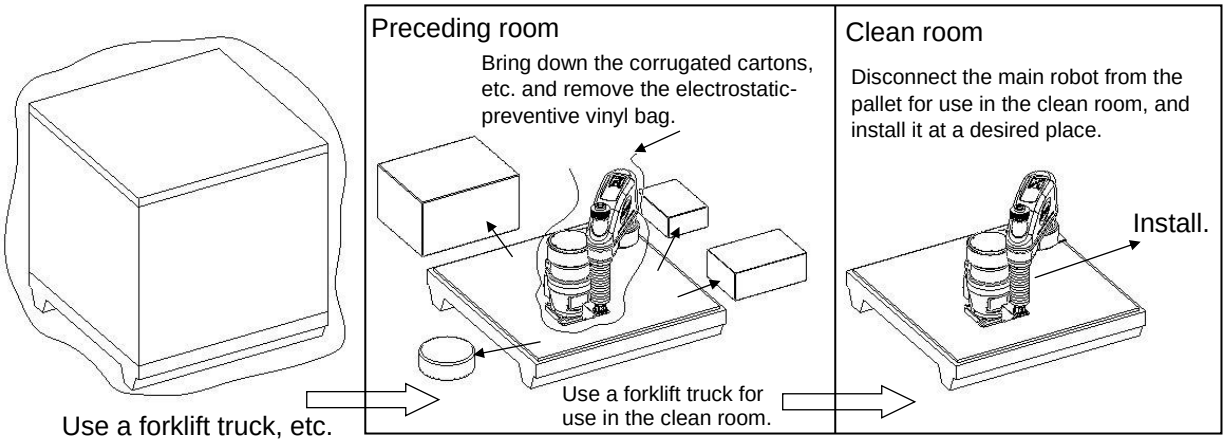



Fig. 2.2 Unpacking procedures

**! CAUTION**

- The clean package level is equivalent to class 100. (Pallet: Made of polypropylene (PP) resin)
- When carrying the robot and controller, take careful precautions not to exert impact or vibration on them.

## 2.2 Mass and Outer Dimensions

The mass and outer dimensions of the robot at the time of transport are shown in Fig. 2.3.

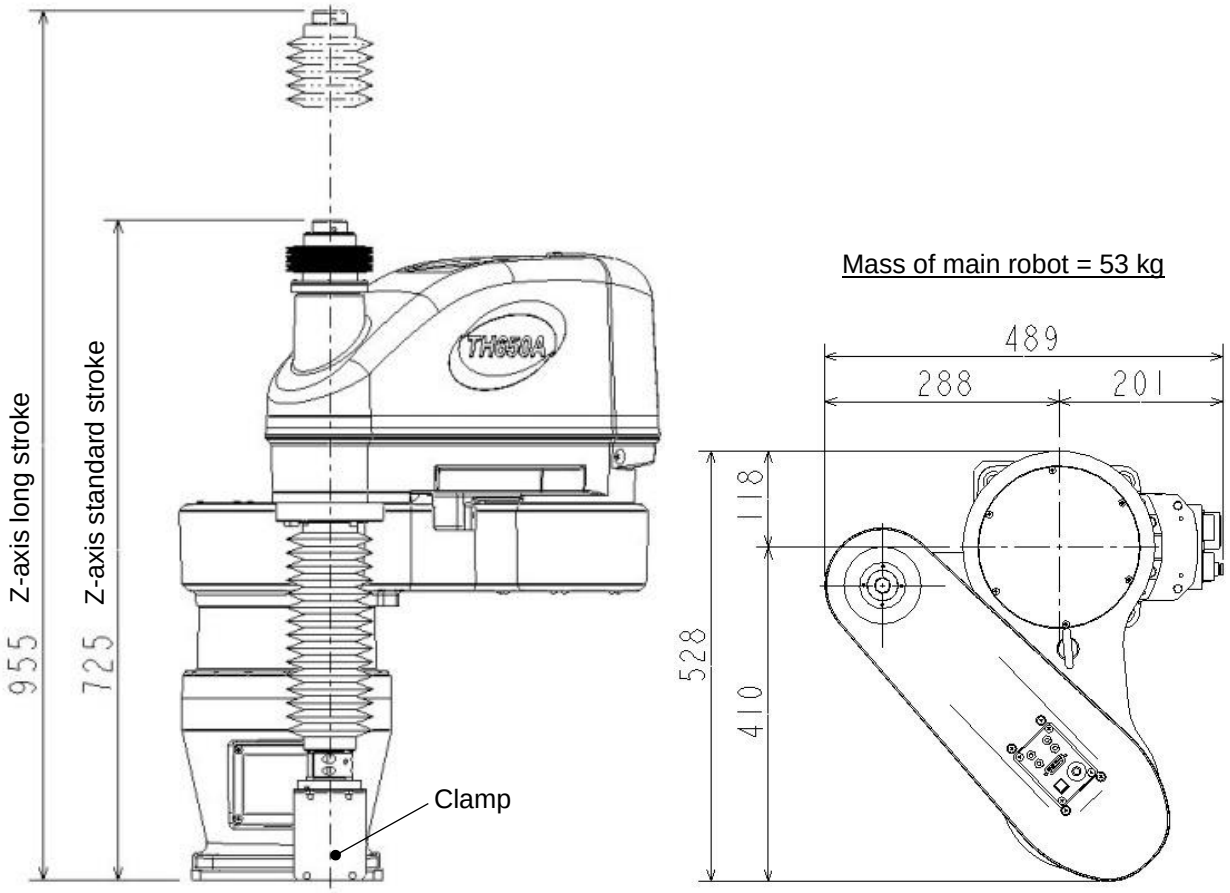



Fig. 2.3 Outer dimensions at transport

## 2.3 Cautions on Transporting the Robot

The clean type robot equals the standard robot added with two (2) bellows. The precautions to be taken at transport are stated below. The precautions other than the below are the same as in the standard robot. See the TH650A Installation and Transportation Manual provided separately.

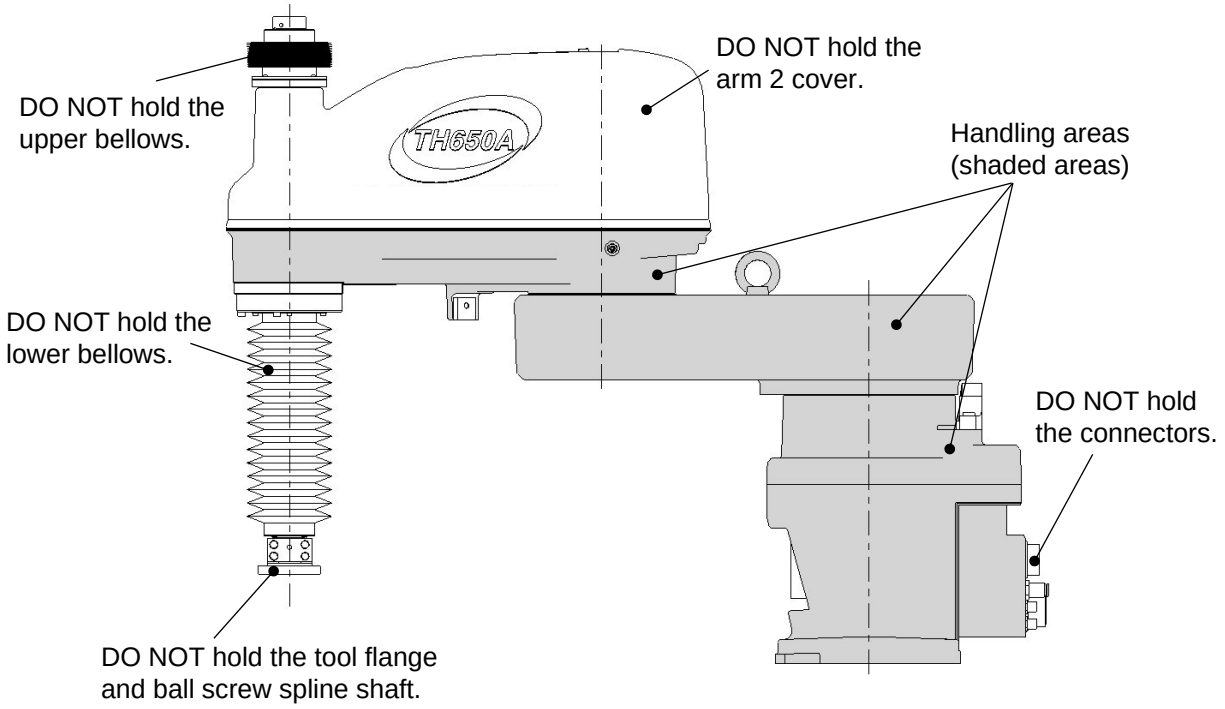



Fig. 2.4 Robot handling prohibited areas

After the installation, remove the clamp and eyebolt used for transport.

**! CAUTION**

- When lifting up the robot by workers, hold the shaded locations by hands, as shown in Fig. 2.4. If the ball screw spline shaft is held by hands, an unusually large force is exerted, resulting in a malfunction.
- If the bellows is held or gripped by hands, it may be broken due to friction with the parts inside.
- When carrying the robot by workers, take careful precautions to prevent their hand or leg from being caught in the robot.
- The work should always be performed by two (2) or more workers.

# 3. Installation

## 3.1 Installation Environment

Table 3.1 shows the environmental conditions for the location in which the robot and controller are to be installed.

Table 3.1 Environmental conditions for robot and controller

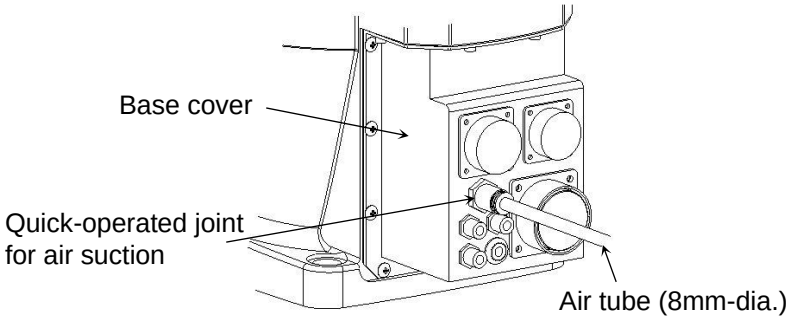
| Item | Specifications |
|---|---|
| Temperature | In operation : 0 to 40°C<br>In storage : –10 to 50°C |
| Humidity | 20 to 90 % (Non-condensing)<br>DO NOT install the robot and controller in a place where they are exposed to water or other liquid. |
| Altitude | 1000 m or less |
| Vibration | In operation : 0.98 $m/s^2$ or less |
| Dust | No inductive dust should exist.<br>When using the robot and controller in a dusty environment, consult with us beforehand. |
| Gas | No corrosive or combustible gas should exist. |
| Sunlight | The robot and controller should not be exposed to direct sunlight. |
| Power noise | A heavy noise source should not exist nearby. |
| Magnetic field | A heavy magnetic field source should not exist nearby. |

**! DANGER**

- Do not place the robot or controller near combustible. Doing so could lead to fires if it ignites due to a fault, etc.

## 3.2 Air Suction Volume

It is possible to maintain the cleanness level of class 10 by sucking a specified volume of air from the quick-operated joint for air suction attached to the base cover (i.e., connector unit of the base).
The air suction unit and suction air tube (8mm-dia.) should be provided by the customer.




| Air suction volume | 60 normal liters/min |
|---|---|
| Class 10 | Ten (10) or less dust particles of 0.1 µm or over in diameter, which exist in 1 cft (28,317 $cm^3$) of a sample area. |

**! CAUTION**

- In the clean room, the down flowrate should be 0.4 m/s or over.
- Unless air suction is executed, dust will generate.

## 3.3 Coordinate System

The robot's joint angle origin (0° or 0 mm position) is factory-calibrated according to the base reference planes. Fig. 3.1 shows the base coordinate system (XB, YB, ZB) and origin of each axis joint angle.

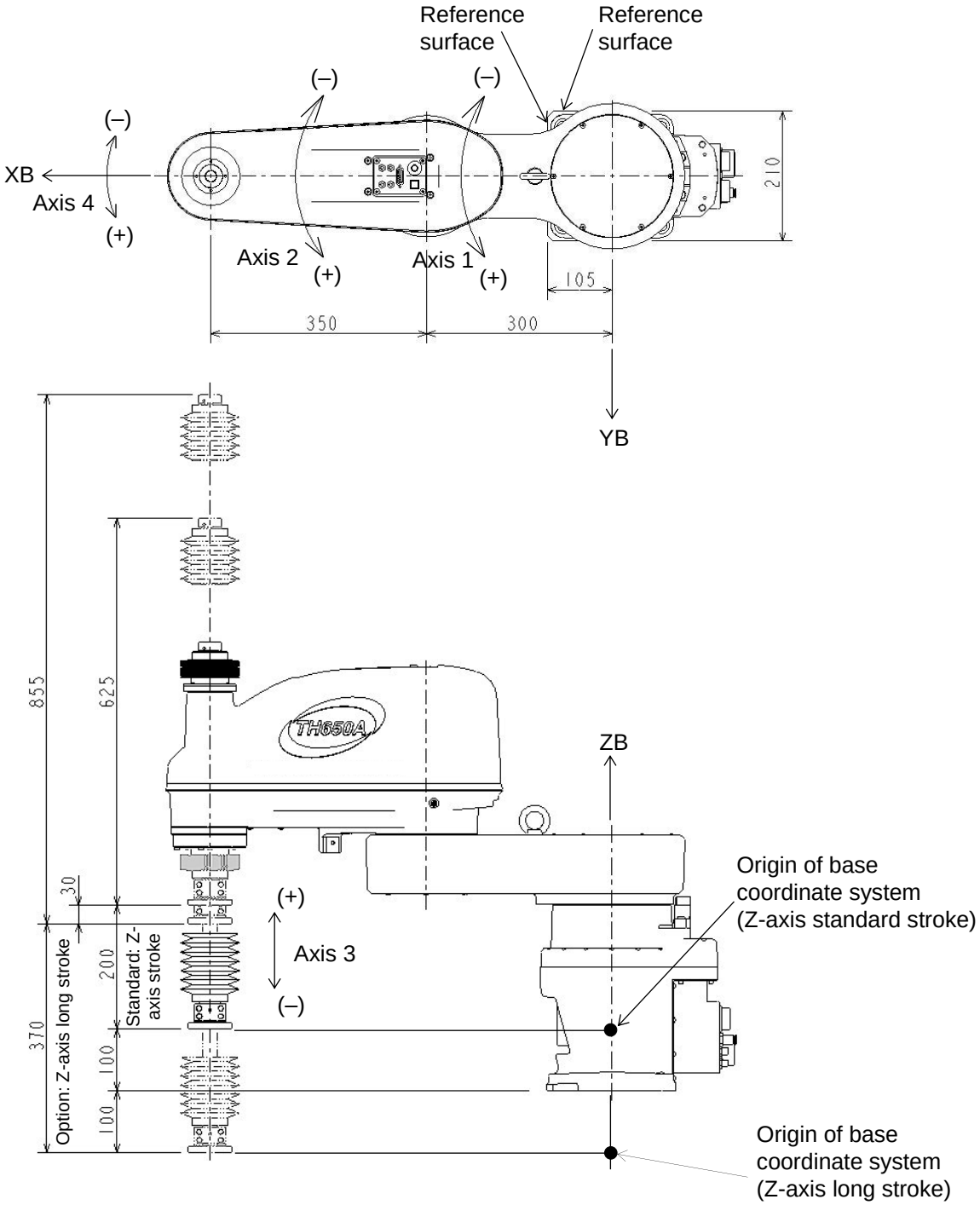



Fig. 3.1 Base coordinate system and joint angle origin

## 3.4 Installing the Robot

The robot is secured, using the set holes on the base (four (4) places).
Use M16 hexagon socket head cap screws (made of stainless steel).
The robot installation method is shown in Fig. 3.2. The reference surfaces are provided on the base.
When you wish to adjust the robot position in the base coordinate system, or replace the robot with another one, provide appropriate reference surfaces and secure the robot by applying them to the reference surfaces of the base.

**! CAUTION**

- The robot will suddenly accelerate and decelerate during operation. When installing it on a frame, make sure that the frame has sufficient strength and rigidity.

  If the robot is installed on a frame that does not have sufficient rigidity, vibration will occur while the robot is operating, and could lead to faults.

  When the robot is installed on the floor, secure it completely with anchor bolts, etc.
- Install the robot on a level place. Failure to do so could lead to a drop in performance or faults.

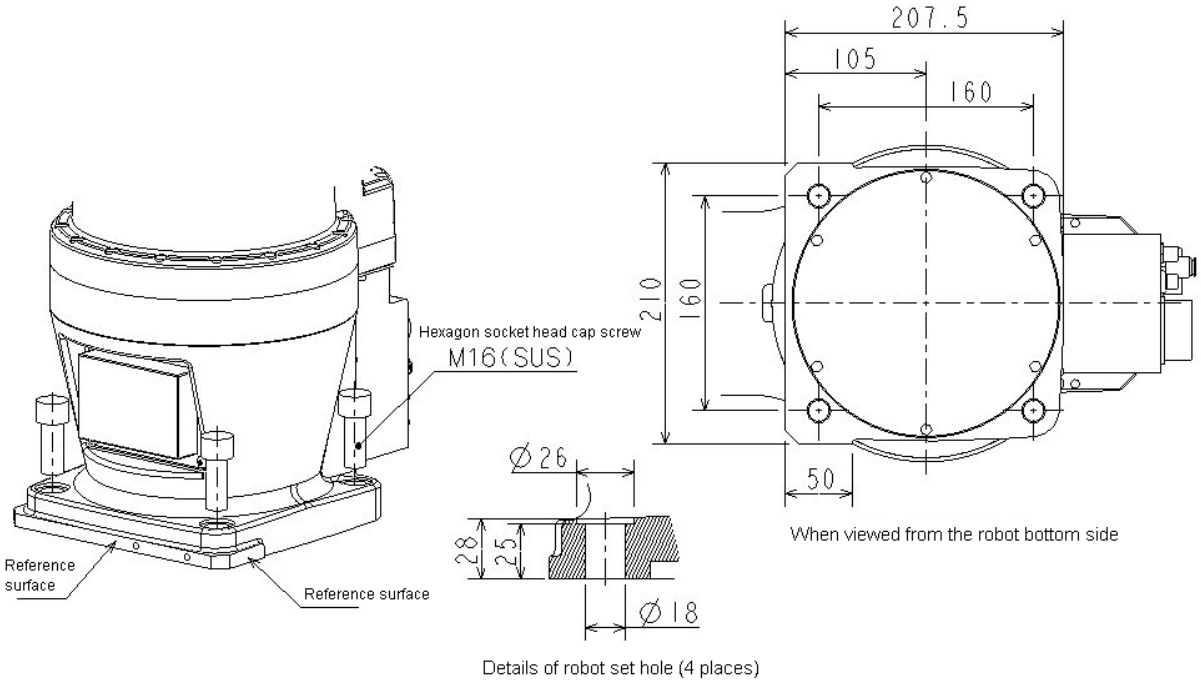


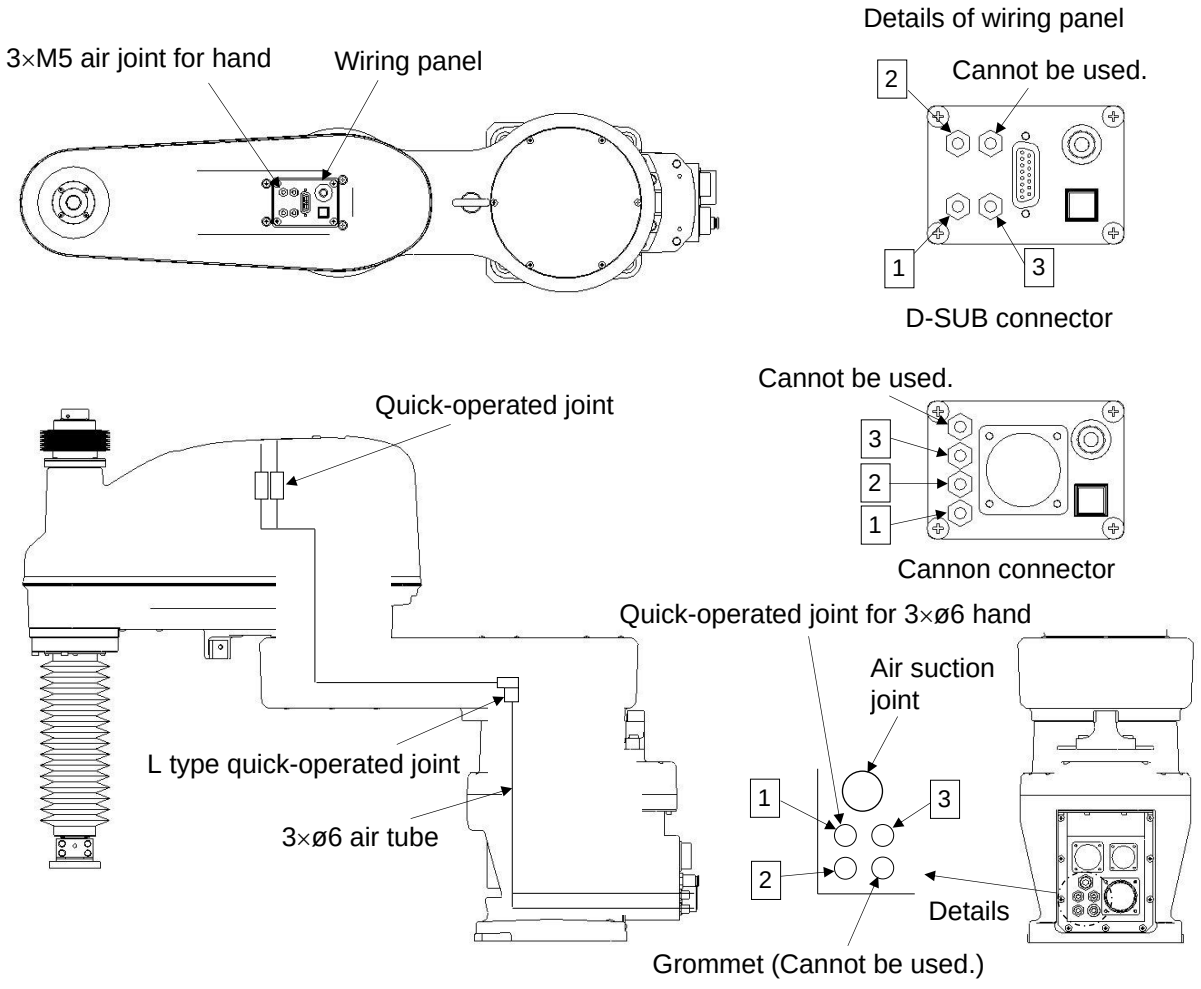

Fig. 3.2 Installation method

## 4. Tool Interface

Mounting of a tool and tool signals are the same as in the standard robot.
For details, see the TH650A Installation and Transportation Manual provided separately.

### 4.1 Tool Air Piping

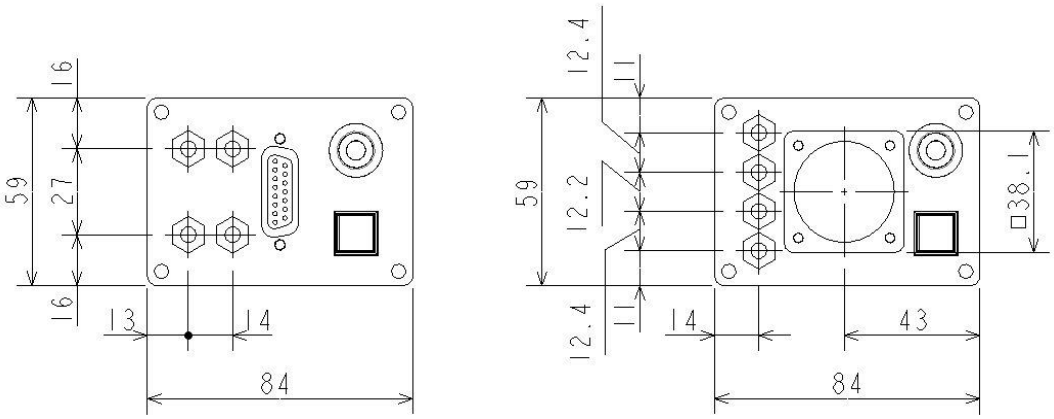
The robot is provided with three (3) air lines for the tool.
The outer diameter of the air pipelines is 6 mm. Fig. 4.1 shows the tool air piping.
The air control unit (oiler, regulator with gage and filter) and solenoid valves should be provided by the user.




The air tube is identified by the number and color. At piping, make sure that each tube is connected properly, referring to the below-mentioned.

1 : Red 2 : White 3 : Blue

Air joint pitches of the panel




D-SUB connector

Cannon connector (option)

Fig. 4.1 Tool air piping

# 5. Maintenance

## 5.1 Items for Maintenance and Inspection

The maintenance schedule and maintenance procedures for the clean type robot are shown in Table 5.1 below.
The basic structure of the clean type robot is the same as that of the standard robot. For the maintenance schedule, maintenance procedures, contents of inspection, etc., see the TH650A Maintenance Manual provided separately.

Table 5.1 Maintenance items for clean type robot

D: Daily inspection Q: Quarterly inspection
S: Semi-annual inspection A: Annual inspection

| Check position | Description | D | Q | S | A | Refer to |
|---|---|---|---|---|---|---|
| Check for bellows | Visually make sure that the upper and lower bellows are not damaged or cracked and in a good shape. | O | O | O | O | Para. 5.1.1 |
| Replacement of bellows | Replace the bellows If it is damaged, or according to the maintenance schedule. | | | | O | Para. 5.1.2 |
| Filling grease to ball screw spline unit | Remove the bellows and fill the grease to the ball screw spline. After the filling, mount the bellows as it was set originally. | | | O | O | Para. 5.1.3 |



**DANGER**

- Be sure to turn off the main power switch of the controller before approaching the robot for maintenance and inspection.



**CAUTION**

- When replacing the bellows with a new one, dust will generate. Before the replacement, therefore, move the robot to a place where dust generation poses no problem at all.

### 5.1.1 Check for Bellows

Visually make sure that each bellows is not found broken, worn out, twisted or folded during robot operation or stop.
If it is found broken or worn out, the cleanness level will drop. Replace it with a new one immediately. If the bellows is twisted or folded, it will be damaged. When this happens, correctly set it again.

### 5.1.2 Replacement of Bellows

Replace the bellows if it is damaged, or according to the maintenance schedule.
For the maintenance schedule, replace the bellows when the axis 3 has shuttled 30 million counts or when one (1) year has passed after the robot operation, whichever comes first.

### 5.1.3 Filling Grease to Ball Screw Spline Unit

Dismantle each bellows, referring to Para. 5.4. Then fill the grease to the ball screw spline unit, referring to the TH650A Maintenance Manual provided separately.

## 5.2 Layout of Robot Components

The layout of the robot mechanical components is shown in Fig. 5.1.

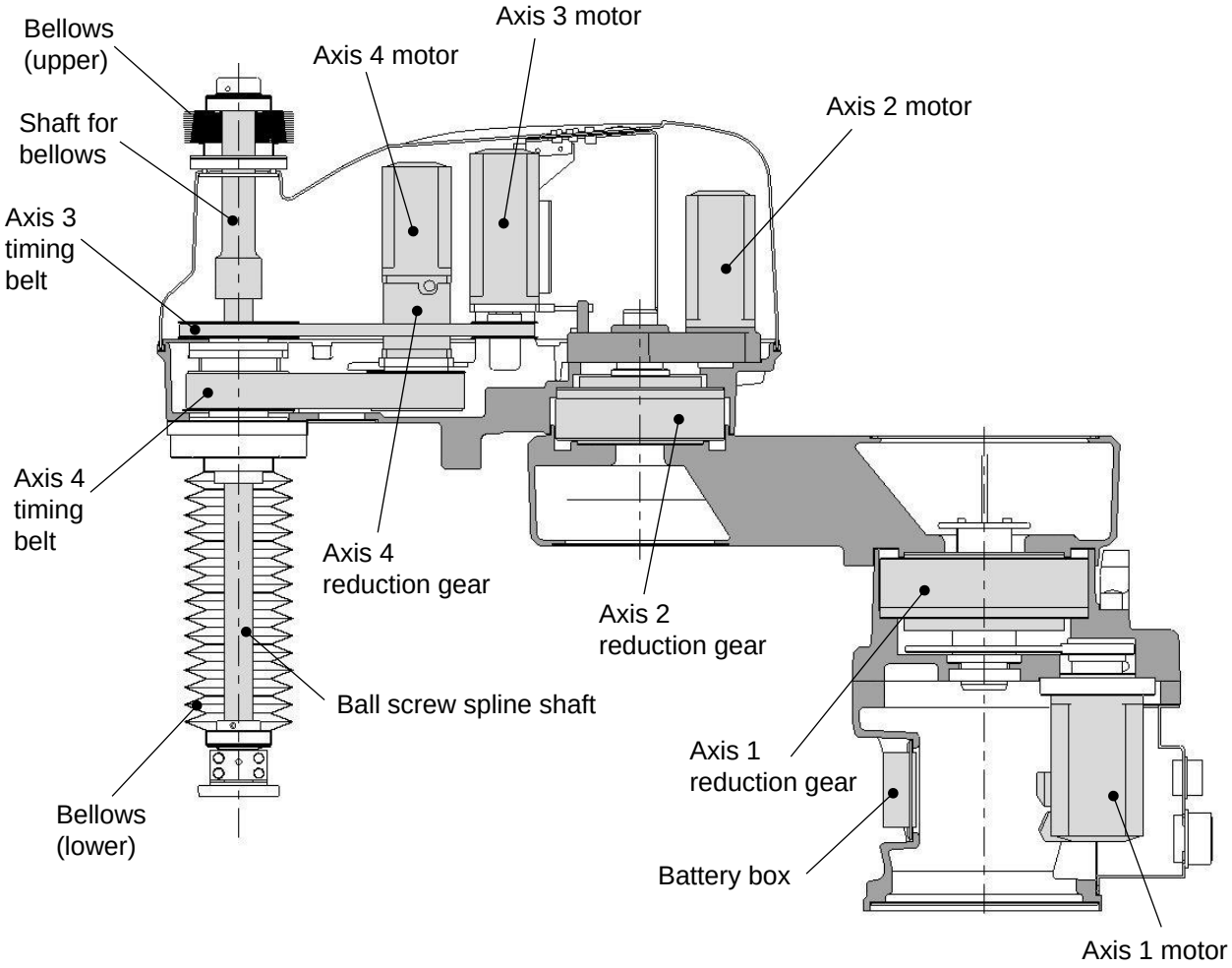



Fig. 5.1 Layout of robot mechanical components

**! CAUTION**

- When replacing the bellows with a new one or when mounting and dismounting the cover, dust will generate. Before starting the work, therefore, move the robot to a place where dust generation poses no problem at all.

## 5.3 Mounting and Dismounting Covers

A gasket (i.e., rubber packing) is attached to each cover set surface of the clean type robot.
Strictly observe the following procedures to mount and dismount each cover.

**! DANGER**

- Before mounting and dismounting each cover, be sure to turn off the main power (“POWER”) switch.
- When opening each cover, make sure that water or contaminant will not enter the robot. If the power is supplied while water or contaminant is left in the robot, you may get an electric shock or the robot may be damaged, which is very dangerous.

### 5.3.1 Base Cover

In all, two (2) base covers are provided; the cover concurrently used for connector panel and the cover concurrently used for battery box.
The replacement procedures for the former cover (i.e., cover concurrently used for connector panel) are the same as those for the standard robot.

The cover concurrently used for battery boxl is secured to the base with four (4) cross-recessed truss head screws (M4 × 6) by inserting the gasket. As the cover is connected with the connectors inside, DO NOT pull out the cover by force.

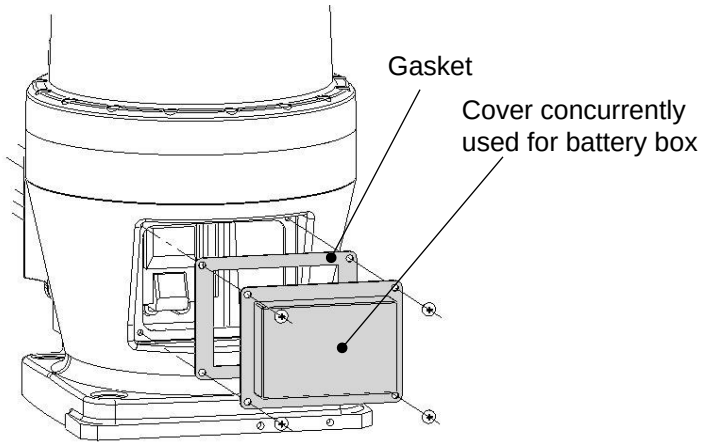



Fig. 5.2 Cover concurrently used for battery box

**CAUTION**

- Be sure to attach the gasket when mounting the cover. Otherwise, the cleanness level will deteriorate.

### 5.3.2 Arm 1 Covers

The arm 1 covers are provided above the axis 1 (arm cover 1) and under the axis 2 (arm cover 2). Each cover is secured to the arm 1 with six (6) cross-recessed truss head screws (M4 × 6) by inserting the gasket.

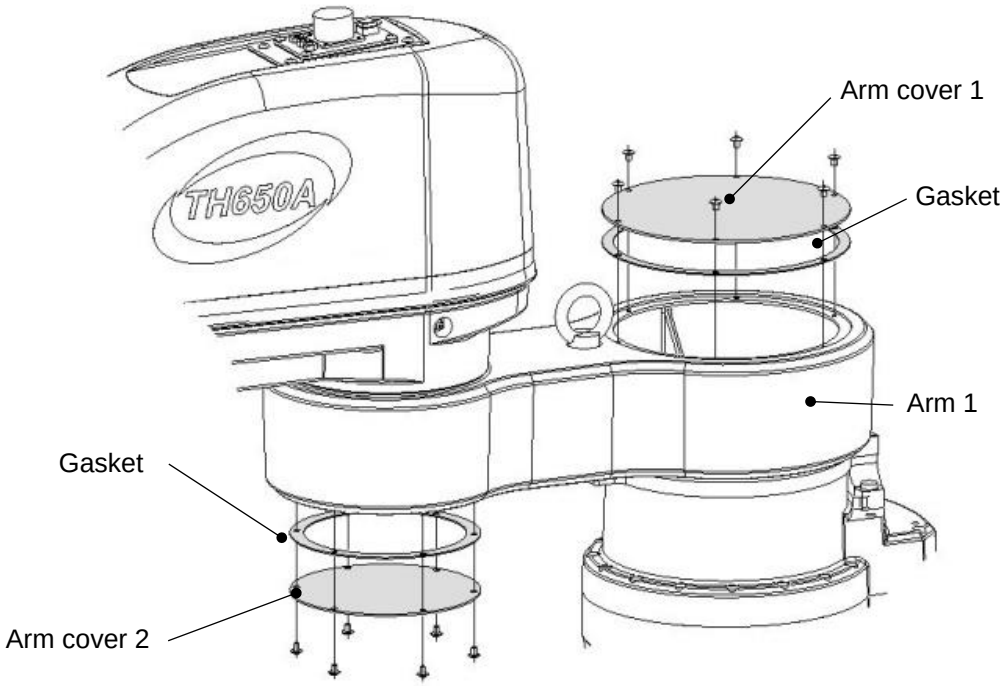



Fig. 5.3 Arm 1 covers

**CAUTION**

- Be sure to attach the gasket when mounting the cover. Otherwise, the cleanness level will deteriorate.

### 5.3.3 Arm 2 Cover

The arm 2 cover is secured to the cover securing bracket with four (4) cross-recessed truss head screws (M4 × 10). Due to the clean type robot, the bellows (upper) is attached to the cover.
To dismount the arm 2 cover, the bellows should be disconnected first. For the replacement procedures for the bellows, see Para. 5.4.
In the descriptions below, the upper bellows is already removed from the cover.

As the cover is embedded to the arm 2, remove the bolts and the cover can be lifted up. It may be hard to disconnect the embedded section. When lifting up the cover, therefore, it is recommended to disconnect the cover by holding the bellows set ring ("A" part of Fig. 5.4). The packing attached to the cover edge can be disconnected together with the cover.

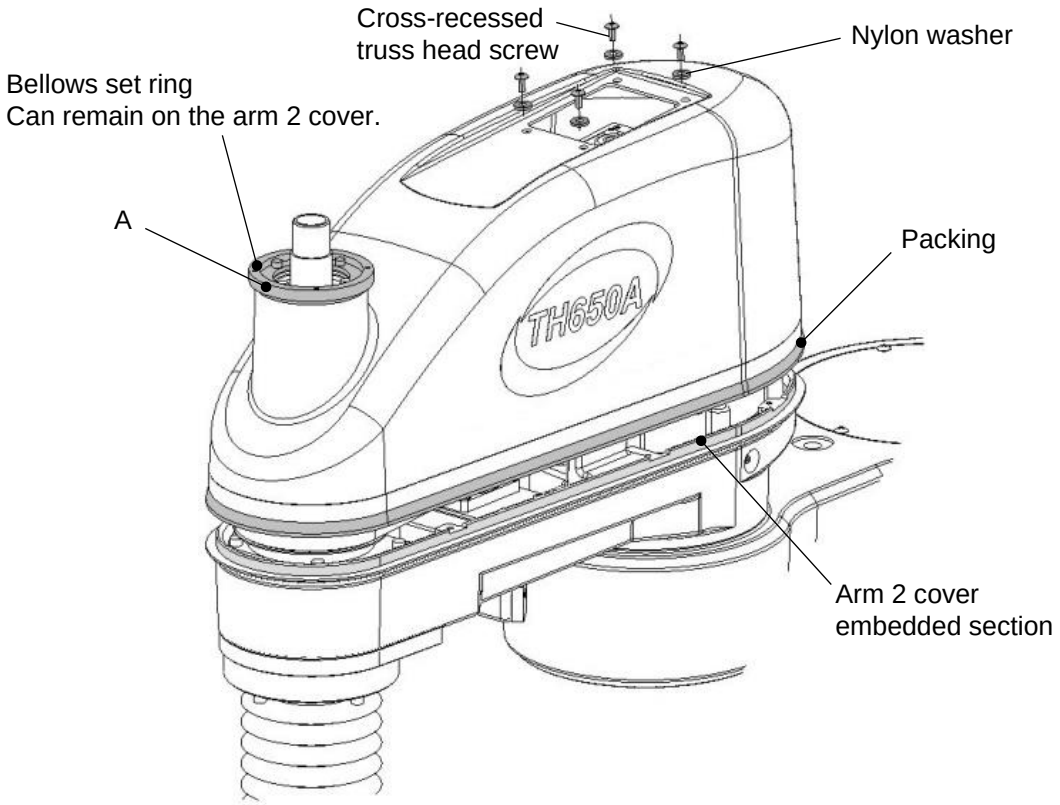



Fig. 5.4 Arm 2 cover

Nylon washers are used to protect the cover. When mounting the cover, be sure to attach the same washers. Also, take careful precautions not to turn up the packing attached to the bottom of the cover.

### 5.3.4 Arm 2 Panel Cover

The cover is provided under the arm 2 so that the wiring panel can be mounted.
For details, see the TH650A Installation and Transportation Manual provided separately.
The cover is secured to the arm 2 with four (4) cross-recessed truss head screws (M4 × 6) by inserting the gasket.
As the gasket is glued to the cover, they can be removed all together.

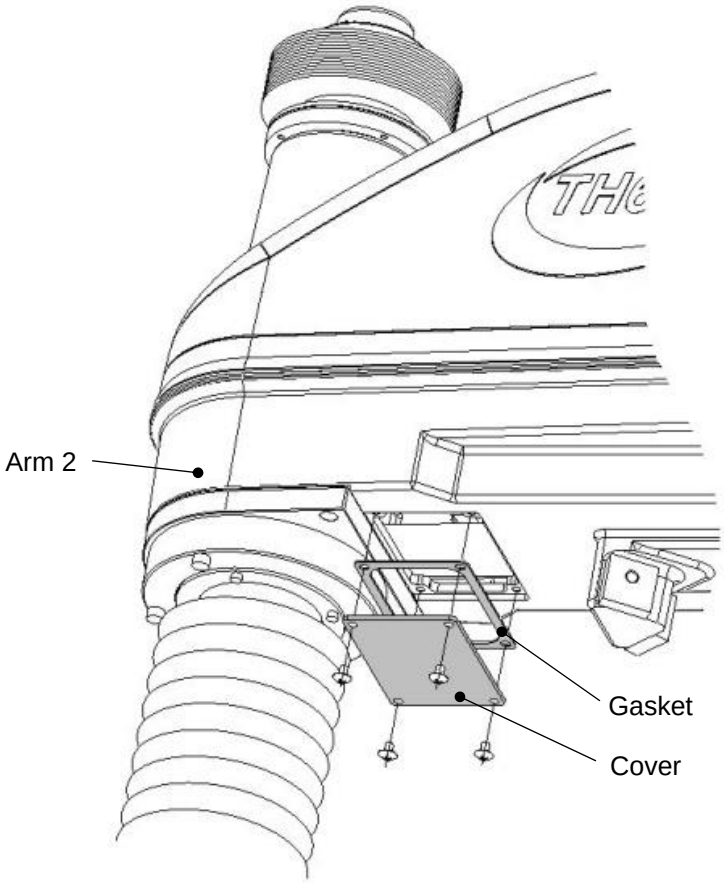



Fig. 5.5 Arm 2 panel cover

**! CAUTION**

- Be sure to attach the gasket when mounting the cover. Otherwise, the cleanness level will deteriorate.

## 5.4 Replacing Bellows

**DANGER**

- Before replacing the bellows with a new one, be sure to turn off the controller power and remove the power plug.

### 5.4.1 Replacement Procedures for Bellows (Lower)

1) Remove the tool flange, followed by the keep plate under the bellows.

2) Remove the keep plate above the bellows, then pull the bellows downward.

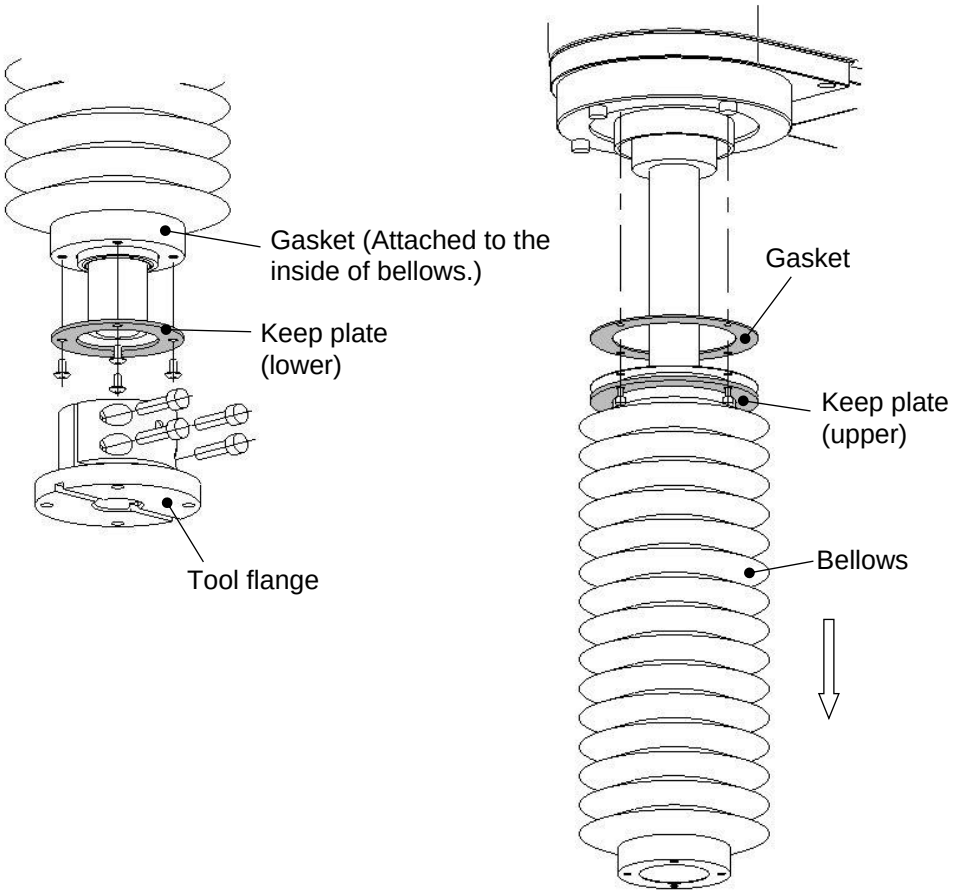



Fig. 5.6 Bellows (lower)

3) To mount the bellows, observe the above steps in the reversed order.
Be sure to attach the gasket.

**! CAUTION**

- When dismantling the bellows, DO NOT pull it out by force. Otherwise, the bellows may rupture.
- Be sure to attach the gasket. Otherwise, the cleanness level will deteriorate.

5.4.2 Replacement Procedures for Bellows (Upper)

1) Remove the three (3) set screws (4 × 6 SUS), then pull out the wiring guide upward.

2) Remove the cross-recessed truss head screws (3 × 6 SUS) securing the bellows. (Four (4) screws each on the top and bottom sides; a total of eight (8) screws) The bellows is secured by the keep plate with the gasket inserted on both the top and bottom sides.

3) Pull out the bellows upward. It can be pulled out even when the bearing case remains attached.

4) To remove the arm 2 cover, pull out the bearing case upward.
The bearing is secured with three (3) cross-recessed truss head screws in the bearing case. The bearing need not be disconnected.

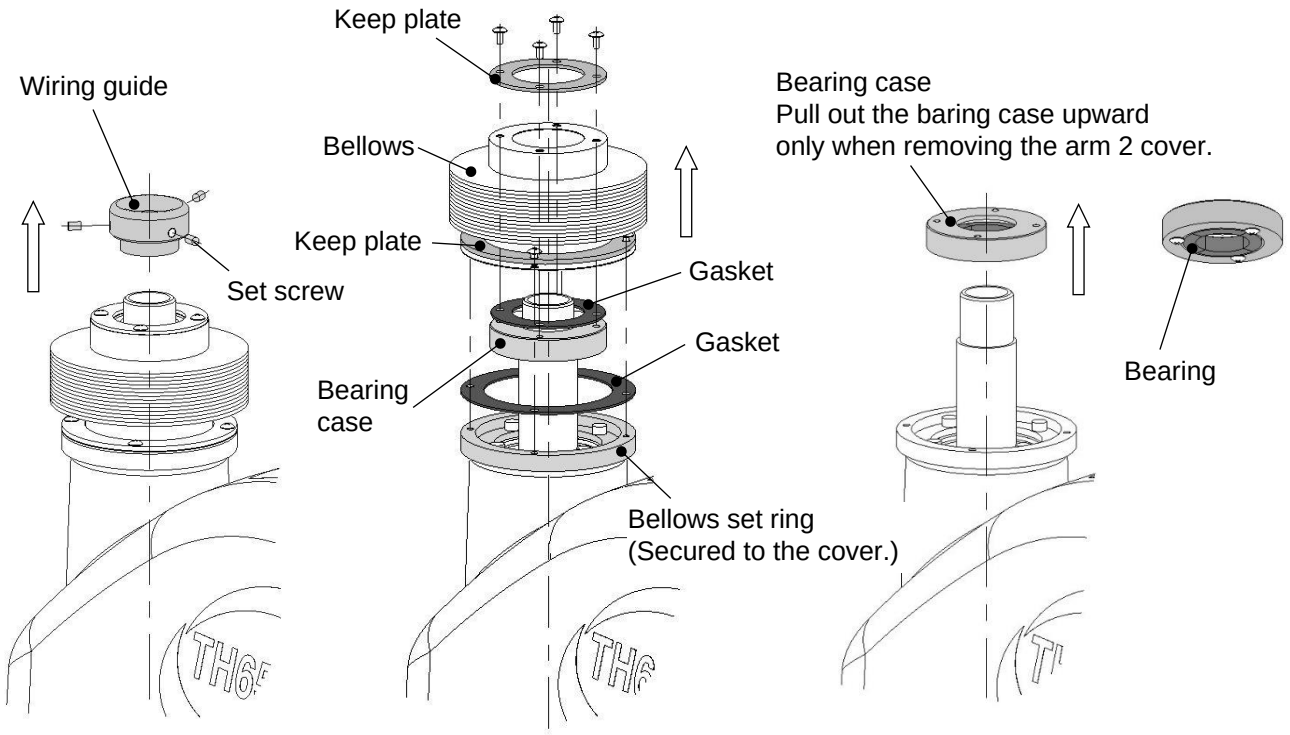



Fig. 5.7 Bellows (upper)

5) To mount the bellows, observe the above steps in the reversed order.
Be sure to attach the gaskets.

**! CAUTION**

- When dismantling the bellows, DO NOT pull it out by force. Otherwise, the bellows may rupture.
- Be sure to attach the gaskets. Otherwise, the cleanness level will deteriorate.

# 6. Replacement Parts for Maintenance

## 6.1 Replacement Parts List for Maintenance

| No. | Part name | Type | Our dwg. No. | Unit code | Maker | Q'ty | Remarks |
|---|---|---|---|---|---|---|---|
| 1 | Bellows | | S828912 | Y610A38Z0 | SHIBAURA MACHINE | 1 | Z-axis standard stroke |
| 2 | Bellows | | S835831 | Y610A38Y0 | SHIBAURA MACHINE | 1 | Z-axis long stroke |

The replacement parts for maintenance other than the above are the same as those of the TH650A robot. For details, see the TH650A Maintenance Manual provided separately.

- When you wish to purchase the replacement parts for maintenance, make sure of the serial number of the main robot and contact us.